

Part 1:

Task 1:

1. 192.168.10.0
2. 192.168.10.255
3. Usable hosts range from 192.168.10.1 -> 192.168.10.254, so there are 254 usable hosts that can connect
4. The subnet mask is 255.255.255.000

Task 2:

Subnet	Subnet Address	CIDR Notation	Usable host range
Smart home devices	192.168.10.0	/26	xx.1->xx.62
Gaming devices	192.168.10.64	/26	xx.65->xx.126
Personal devices	192.168.10.128	/26	xx.129->xx.190
Guest devices	192.168.10.192	/26	xx.193->xx.254

Task 3:

Destination Network	Prefix	Next Hop
Smart home devices	192.168.10.0/26	Router Interface 1
Gaming devices	192.168.10.64/26	Router Interface 2
Personal devices	192.168.10.128/26	Router Interface 3
Guest devices	192.168.10.192/26	Router Interface 4
All other traffic	0.0.0.0/0	Internet Gateway

1. A possible source IP address is 192.168.10.12 for the smart home device. We could assume that the data is being sent to a personal device to monitor, meaning a possible IP destination is 192.168.10.135 and packets will be sent to Router Interface 3.
2. A possible source IP address is 192.168.10.68 for the gaming console. Since it is connected to an online server, this falls in to the "All other traffic" bucket meaning the packets will be sent to the internet gateway
3. A possible source IP address is 192.168.10.135 for the personal device. Since it is accessing a public server, this falls in to the "All other traffic" bucket meaning the packets will be sent to the internet gateway.

Task 4:

1. In destination-based IP forwarding, a more specific route can be implemented by adding an entry with a longer prefix length. This also means we increase the CIDR number. The router will try to match this IP first, so a smart lock would be routed to the new interface first instead of going to the more generic smart home devices route. We will call the interface “Security Gateway”
2. Yes, the packet could match multiple entries. It could match the most specific entry (for only the smart lock), the generic smart home devices entry, and the default route.
3. The routing logic would choose the new, specific route when traffic sources from the smart lock. The selection logic is based on the number of bits that match. Because the new route has more bits, meaning more possible bits to match, it will be chosen over any more generic route. The Security Gateway interface is chosen due to this.
- 4.

Destination Network	Prefix	Next Hop
Smart lock	192.168.10.5/32	Security Gateway
Smart home devices	192.168.10.0/26	Router Interface 1
Gaming devices	192.168.10.64/26	Router Interface 2
Personal devices	192.168.10.128/26	Router Interface 3
Guest devices	192.168.10.192/26	Router Interface 4
All other traffic	0.0.0.0/0	Internet Gateway

Task 5:

1. Situations:
 - a. UDP Gaming packet: The router will see a couple of key things. First, it will see that the protocol is UDP. It can then also use the MAC address, IP address, and/or source device info to confirm that it is coming from a gaming device. With the packet being a confirmed UDP gaming packet, it can be prioritized for processing and forwarding.
 - b. Email packer: The router will look to confirm that the source port matches a known, common email port (ie 25 for SMTP). It may also look to confirm that the packet is TCP. With these aspects confirmed, the router can “route it differently”, whatever that may mean for the network.
 - c. Traffic from blocked device: The router should mainly use the MAC address and source device info to check whether the device is blocked. The source IP address can also help, but may have been modified from the original blocked

IP, so it should not be the sole determiner. When a blocked device is detected, the packet will be dropped.

2. In this network, matching is first completed by examining the following:

- Who/what sent the packet? (Source devices, MAC, IP)
- Where is it going? (Destination MAC, IP)
- What form is the packet in? (UDP/TCP/ICMP)
- What application is it from? (Source port number)
- What application is it going to? (Destination port number)

Based on the answers to these questions, the router can match the packet to an action. The actions include:

- Dropping packets (blocked device)
- Prioritize/place at front of queue (gaming device)
- Forward (email)

3. The match + action in this situation differs from destination-based forwarding by adding logic outside of IP address to determine what happens to traffic. Because of this, custom rules and actions are much easier to configure. There is also a higher level of security and flexibility.

Part 2:

Task 1:

1. Ipconfig

```

PS C:\Users\jesse> ipconfig

Windows IP Configuration

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):
```

Connection-specific DNS Suffix	:	
Link-local IPv6 Address	:	fe80::2675:6aa:f384:72db%59
IPv4 Address.	:	172.29.16.1
Subnet Mask	:	255.255.240.0
Default Gateway	:	

```

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : lan
IPv6 Address. . . . . : 2600:6c5a:47f0:1810::17a8
IPv6 Address. . . . . : 2600:6c5a:47f0:1810:3c2d:6f3a:8536:992c
Temporary IPv6 Address. . . . . : 2600:6c5a:47f0:1810:d87d:7255:d500:52f1
Link-local IPv6 Address . . . . . : fe80::737e:5903:b7c:df8f%12
IPv4 Address. . . . . : 192.168.1.5
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::bed5:edff:fe03:3c02%12
                             192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

```

2. Ipconfig /release

```

PS C:\Users\jesse> ipconfig /release

Windows IP Configuration

No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 3 while it has its media disconnected.
No operation can be performed on Local Area Connection* 4 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::2675:6aa:f384:72db%59
    IPv4 Address. . . . . : 172.29.16.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Unknown adapter Local Area Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : 2600:6c5a:47f0:1810::17a8
    IPv6 Address. . . . . : 2600:6c5a:47f0:1810:3c2d:6f3a:8536:992c
    Temporary IPv6 Address. . . . . : 2600:6c5a:47f0:1810:d87d:7255:d500:52f1
    Link-local IPv6 Address . . . . . : fe80::737e:5903:b7c:df8f%12
    Default Gateway . . . . . : fe80::bed5:edff:fe03:3c02%12

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

```

3. Ipconfig /renew

```

PS C:\Users\jesse> ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection 2 while it has its media disconnected.
No operation can be performed on Local Area Connection* 3 while it has its media disconnected.
No operation can be performed on Local Area Connection* 4 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):
```

Connection-specific DNS Suffix	:	:
Link-local IPv6 Address	:	fe80::2675:6aa:f384:72db%59
IPv4 Address.	:	172.29.16.1
Subnet Mask	:	255.255.240.0
Default Gateway	:	:

```

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  :

Unknown adapter Local Area Connection 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  :

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  :

Wireless LAN adapter Local Area Connection* 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  : lan
IPv6 Address. . . . . : 2600:6c5a:47f0:1810::17a8
IPv6 Address. . . . . : 2600:6c5a:47f0:1810:3c2d:6f3a:8536:992c
Temporary IPv6 Address. . . . . : 2600:6c5a:47f0:1810:d87d:7255:d500:52f1
Link-local IPv6 Address . . . . . : fe80::737e:5903:b7c:df8f%12
IPv4 Address. . . . . : 192.168.1.5
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::bed5:edff:fe03:3c02%12
                           192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  :

```

4. Ipconfig

```

PS C:\Users\jesse> ipconfig

Windows IP Configuration

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):
```

Connection-specific DNS Suffix	:	
Link-local IPv6 Address	:	fe80::2675:6aa:f384:72db%59
IPv4 Address.	:	172.29.16.1
Subnet Mask	:	255.255.240.0
Default Gateway	:	

```

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Unknown adapter Local Area Connection 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:
```

Connection-specific DNS Suffix	:	lan
IPv6 Address.	:	2600:6c5a:47f0:1810::17a8
IPv6 Address.	:	2600:6c5a:47f0:1810:3c2d:6f3a:8536:992c
Temporary IPv6 Address.	:	2600:6c5a:47f0:1810:d87d:7255:d500:52f1
Link-local IPv6 Address	:	fe80::737e:5903:b7c:df8f%12
IPv4 Address.	:	192.168.1.5
Subnet Mask	:	255.255.255.0
Default Gateway	:	fe80::bed5:edff:fe03:3c02%12
	:	192.168.1.1

```

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

PS C:\Users\jesse>

```

Task 2:

1. Discover

```

Frame 632: Packet, 346 bytes on wire (2768 bits), 346 bytes captured (2768 bits) on interface \Device\NPF_{68BA0143-E92D-42B2-B83F-CA53B5F3F147}, id 6
Ethernet II, Src: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 332
  Identification: 0xe22f (57903)
  0000 .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: UDP (17)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 0.0.0.0
  Destination Address: 255.255.255.255
  [Stream index: 13]
User Datagram Protocol, Src Port: 68, Dst Port: 67
  Source Port: 68
  Destination Port: 67
  Length: 312
  Checksum: 0xc3c3 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 58]
  [Stream Packet Number: 1]
  [Timestamps]
  UDP payload (304 bytes)
Dynamic Host Configuration Protocol (Discover)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x84d6dd5e
  Seconds elapsed: 0
  Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  Option: (53) DHCP Message Type (Discover)
  Option: (61) Client identifier
  Option: (50) Requested IP Address (192.168.1.5)
    Length: 4
    Requested IP Address: 192.168.1.5
  Option: (12) Host Name
  Option: (60) Vendor class identifier

```

2. Offer

```

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.5
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  Total Length: 334
  Identification: 0x0d09 (3337)
  0000 .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: UDP (17)
  Header Checksum: 0xe87f [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.1
  Destination Address: 192.168.1.5
  [Stream index: 12]
User Datagram Protocol, Src Port: 67, Dst Port: 68
  Source Port: 67
  Destination Port: 68
  Length: 314
  Checksum: 0x9dc2 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 19]
  [Stream Packet Number: 2]
  [Timestamps]
  UDP payload (306 bytes)
Dynamic Host Configuration Protocol (Offer)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x84d6dd5e
  Seconds elapsed: 0
  Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.1.5
  Next server IP address: 192.168.1.1
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  Option: (53) DHCP Message Type (Offer)
  Option: (54) DHCP Server Identifier (192.168.1.1)
  Option: (61) Client identifier
  Option: (51) IP Address Lease Time
  Option: (58) Renewal Time Value
  Option: (59) Rebinding Time Value
  Option: (1) Subnet Mask (255.255.255.0)
  Option: (28) Broadcast Address (192.168.1.255)
  Option: (15) Domain Name
  Option: (6) Domain Name Server
  Option: (3) Router
  Option: (255) End

```


3. Request

```

> Frame 871: Packet, 374 bytes on wire (2992 bits), 374 bytes captured (2992 bits) on interface \Device\NPF_{68BA0143-E92D-42B2-B83F-CAS3B5F3F147}, id 0
> Ethernet II, Src: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total length: 360
    Identification: 0xe230 (57904)
  > 0000 .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 0.0.0.0
    Destination Address: 255.255.255.255
    [Stream index: 13]
> User Datagram Protocol, Src Port: 68, Dst Port: 67
  Source Port: 68
  Destination Port: 67
  Length: 340
  Checksum: 0x0074 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 58]
  [Stream Packet Number: 2]
  > [Timestamps]
  > UDP payload (332 bytes)
> Dynamic Host Configuration Protocol (Request)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x84d6dd5e
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Options (63) DHCP Message Type (Request)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.1.5)
  > Option: (54) DHCP Server Identifier (192.168.1.1)
  > Option: (12) Host Name
  > Option: (81) Client Fully Qualified Domain Name
  > Option: (60) Vendor class identifier
  > Option: (55) Parameter Request List
  > Option: (255) End

```

4. Acknowledge

```

> Frame 872: Packet, 378 bytes on wire (3024 bits), 378 bytes captured (3024 bits) on interface \Device\NPF_{68BA0143-E92D-42B2-B83F-CAS3B5F3F147}, id 0
> Ethernet II, Src: SagcomBroa_03:3c:02 (bc:d5:ed:03:3c:02), Dst: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.5
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total length: 360
    Identification: 0x0d0a (3338)
  > 0000 .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: UDP (17)
    Header Checksum: 0xe864 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.1
    Destination Address: 192.168.1.5
    [Stream index: 12]
> User Datagram Protocol, Src Port: 67, Dst Port: 68
  Source Port: 67
  Destination Port: 68
  Length: 340
  Checksum: 0xea91 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 19]
  [Stream Packet Number: 3]
  > [Timestamps]
  > UDP payload (332 bytes)
> Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x84d6dd5e
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.1.5
  Next server IP address: 192.168.1.1
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_f3:f1:a1 (dc:46:28:f3:f1:a1)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (ACK)
  > Option: (54) DHCP Server Identifier (192.168.1.1)
  > Option: (61) Client identifier
  > Option: (51) IP Address Lease Time
  > Option: (58) Renewal Time Value
  > Option: (59) Rebinding Time Value
  > Option: (1) Subnet Mask (255.255.255.0)
  > Option: (28) Broadcast Address (192.168.1.255)
  > Option: (15) Domain Name
  > Option: (81) Client Fully Qualified Domain Name
  > Option: (6) Domain Name Server
  > Option: (3) Router
  > Option: (255) End

```

1.

Stage	Source IP	Destination IP	Explanation
Discover	0.0.0.0	255.255.255.255	Client joins network with no IP Address. It sends out a message to search for a DHCP server.
Offer	192.168.1.1	192.168.1.5	DCHP server receives message and searches for available IP addresses. One is sent back to the client
Request	0.0.0.0	255.255.255.255	The client accepts the IP address, and broadcasts this acceptance
Acknowledge	192.168.1.1	192.168.1.5	DCHP server confirms the IP address has been set, and the client is now fully connected to the network.

2. The DHCP server's IP is 192.168.1.1 and the port number is 67